

- Path cost $g(n) = 1$ per move
- Heuristic $h(n)$ = number of misplaced tiles (excluding blank)
- $f(n) = g(n) + h(n)$

Initial State

Code

```
1 2 3
4 _ 6
7 8 5
```

Goal:

Code

```
1 2 3
4 5 6
7 8 _
```

Misplaced tiles in start = {5} $\rightarrow h = 1$

A* Iteration-wise Table

Iteration 0 (Start Node)

Iter	State	$g(n)$	$h(n)$	$f(n)$
0	1 2 3 / 4 _ 6 / 7 8 5	0	1	1

Iteration 1 – Expand Start

Possible moves: Up, Down, Left, Right

Move	State	g	h (misplaced)	$f=g+h$
Up	1 _ 3 / 4 2 6 / 7 8 5	1	2 ({2,5})	3
Down	1 2 3 / 4 8 6 / 7 _ 5	1	2 ({8,5})	3
Left	1 2 3 / _ 4 6 / 7 8 5	1	2 ({4,5})	3
Right	1 2 3 / 4 6 _ / 7 8 5	1	2 ({6,5})	3

Tie $\rightarrow$ choose **Right** (standard tie-breaking)

Iteration 2 – Expand (Right)

State:

Code

```
1 2 3
4 6 _
7 8 5
```

$g = 1, h = 2, f = 3$

Successors:

Move	State	g	h	f
Up	1 2 _ / 4 6 3 / 7 8 5	2	3 ({3,6,5})	5
Down	1 2 3 / 4 6 5 / 7 8 _	2	2 ({6,5})	4
Left	1 2 3 / 4 _ 6 / 7 8 5	2	1 ({5})	3

Best node (lowest f): Left → back to initial pattern but with higher g

Next best unexplored: Down (f=4)

Iteration 3 – Expand (Down)

State:

Code

```
1 2 3
4 6 5
7 8 _
```

$g = 2, h = 2, f = 4$

Successors:

Move	State	g	h	f
Up	1 2 3 / 4 6 _ / 7 8 5	3	2	5
Left	1 2 3 / 4 6 5 / 7 _ 8	3	3 ({6,5,8})	6

Lowest f remains previous frontier nodes with f=3 already expanded, so continue.

Optimal Path Found by A*

(Using lowest cumulative evaluation)

Code

Start

1 2 3

4 _ 6

7 8 5

→ Right

1 2 3

4 6 _

7 8 5

→ Down

1 2 3

4 6 5

7 8 _

→ Left

1 2 3

4 5 6

7 8 _

Goal reached.

Final Optimal Path Cost

Total moves = 3
